Jinkun Geng, Ph.D.

Building and verifying efficient and resilient systems.

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Employment History

2026.01 – Present — **Assistant Professor/Faculty Fellow,** New York University. Formal verification of sequencer-aided coordination (SAC) protocols, with Professors Anirudh Sivaraman, Joseph Tassarotti, and Shuai Mu.

2025.06 – 2026.01 — **Senior Postdoctoral Associate,** Stony Brook University. Geo-distributed transaction processing (Tiga), with Professors Shuai Mu and Anirudh Sivaraman.

2024.01 – 2025.05 — **Senior Software Researcher,** Clockwork Systems, Inc. NCCL flow scheduling and load balancing for LLMs (FlowRocket).

2022.06 – 2022.09 — **Software Engineer Intern,** Clockwork Systems, Inc. Optimized and open-sourced Nezha consensus.

2020.06 – 2020.09 — **Software Engineer Intern,** Google. Video encoding optimization: warped motion vector prediction and DBSCAN clustering of motion vectors.

Education

2019 – 2024 — **Ph.D., Stanford University** Computer Science
Thesis title: *Accelerating Distributed Protocols with Clock Synchronization*
Advisors: Professors Balaji Prabhakar, Mendel Rosenblum, and Anirudh Sivaraman

2016 – 2019 — **M.Sc., Tsinghua University** Computer Science
Thesis title: *Research on Performance Optimization of Large-Scale Matrix Factorization*
Advisor: Professor Dan Li

2012 – 2016 — **B.Eng., Beihang University** Software Engineering

Projects

System Formal Verification, 2025.06 – Present — Implementing and verifying a broad family of Sequencer-Aided Coordination protocols, including NOPaxos, Corfu, LazyLog, etc. Our project pairs high-performance implementations with machine-checked proofs of correctness.

Tiga, 2023 – 2025 — Designed and implemented Tiga, a geo-distributed transaction protocol that uses synchronized clocks to consolidate consensus and concurrency control. Presented at SOSP'25 (video talk); invited talk at Azure Databases Tech Talk Forum (link).

FlowRocket, 2024 – 2025.05 — Developed flow scheduling and load balancing algorithms on top of NCCL to accelerate LLM workloads. The work resulted in two patents and was integrated into Clockwork's commercial products (link).

Nezha, 2021 – 2024 — Designed Nezha, a deployable, high-performance consensus protocol with synchronized clocks (10× higher throughput than Raft); open-sourced at Clockwork Systems (code). Presented at VLDB (video talk); studied by MongoDB researchers (link).

CloudEx, 2019 – 2024 — Built CloudEx, a fair-access financial exchange in the public cloud (HotOS'21), and developed network support for scalable, high-performance cloud exchanges (Onyx, SIGCOMM'25). CloudEx served as the case study for the SIGCOMM'20 NetFinance tutorial (link).

RDMA for DML, 2017 – 2019 — Applied RDMA to accelerate distributed machine learning (DML) workloads, with results published in multiple top-tier venues.

Selected Publications

1. **Jinkun Geng**, Muhammad Haseeb, Daniel Qian, Joseph Tassarotti, Shuai Mu, and Anirudh Sivaraman, "Formal Linearizability Proof for Sequencer-Aided Coordination Protocols," Under submission, 2026.
2. **Jinkun Geng**, Shuai Mu, Anirudh Sivaraman, and Balaji Prabhakar, "Tiga: Accelerating Geo-Distributed Transactions with Synchronized Clocks," in *Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles*, ser. SOSP '25, 2025, pp. 555–571. DOI: 10.1145/3731569.3764854.
3. Muhammad Haseeb, **Jinkun Geng**, Daniel Duclos-Cavalcanti, Xiyu Hao, Ulysses Butler, Radhika Mittal, Srinivas Narayana, and Anirudh Sivaraman, "Network Support for Scalable and High Performance Cloud Exchanges," in *Proceedings of the ACM SIGCOMM 2025 Conference*, ser. SIGCOMM '25, 2025, pp. 1110–1131. DOI: 10.1145/3718958.3750530.
4. **Jinkun Geng**, Anirudh Sivaraman, Balaji Prabhakar, and Mendel Rosenblum, "Nezha: Deployable and High-Performance Consensus Using Synchronized Clocks," *Proc. VLDB Endow.*, vol. 16, no. 4, pp. 629–642, Dec. 2022. DOI: 10.14778/3574245.3574250.
5. Ahmad Ghalayini, **Jinkun Geng**, Vighnesh Sachidananda, Vinay Sriram, Yilong Geng, Balaji Prabhakar, Mendel Rosenblum, and Anirudh Sivaraman, "CloudEx: A Fair-Access Financial Exchange in the Cloud," in *Proceedings of the Workshop on Hot Topics in Operating Systems*, ser. HotOS '21, 2021, pp. 96–103. DOI: 10.1145/3458336.3465278.
6. Shuai Wang, **Jinkun Geng**, and Dan Li, "Impact of Synchronization Topology on DML Performance: Both Logical Topology and Physical Topology," **(co-first author)**, *IEEE/ACM Transactions on Networking*, vol. 30, no. 2, pp. 572–585, Oct. 2021. DOI: 10.1109/TNET.2021.3117042.
7. **Jinkun Geng**, Dan Li, and Shuai Wang, "Rima: An RDMA-Accelerated Model-Parallelized Solution to Large-Scale Matrix Factorization," in *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, 2019, pp. 100–111. DOI: 10.1109/ICDE.2019.00018.
8. **Jinkun Geng**, Dan Li, Yang Cheng, Shuai Wang, and Junfeng Li, "HiPS: Hierarchical Parameter Synchronization in Large-Scale Distributed Machine Learning," in *Proceedings of the 2018 Workshop on Network Meets AI & ML*, ser. NetAI'18, 2018, pp. 1–7. DOI: 10.1145/3229543.3229544.

Teaching Experience

Instructor	NYU CSCI-UA.0480-062: Computer Networks, Fall 2026
	NYU INDIV-UG 1903: Independent Study, Fall 2026
	NYU CSCI-UA.0480-062: Computer Networks, Spring 2026
Teaching Assistant	Stanford CS 251: Blockchain Technologies
	Stanford CS 349F: Technology for Financial Systems

Professional Service

2026	Technical Program Committee (TPC) Member, ACM SIGCOMM 2026

Honors and Awards

2026	NYU Faculty Research & Professional Development Fund ($2,500)
2025	Google Research Credit Award ($5,000)
2019	Outstanding Master Graduate at Tsinghua University (3 out of 141)
	Outstanding Master Dissertation at Tsinghua University (7 out of 141)
2014	Samsung Enterprise Scholarship